JAN-D-511
4 DECEMBER 1947

JOINT ARMY-NAVY SPECIFICATION

DYES, GREEN AND SCARLET
(For Use in Ammunition)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective departments and supersedes the following specifications:

U. S. Army
27–2A
1 April 1938
27–3
23 May 1934

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION.

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATION

JAN–C–539—Cloth, Silk, Cartridge-Bag.

U. S. ARMY SPECIFICATIONS

3–1—Paints and Related Materials, General Specifications for Inspection and Tests (Color Card Supplement Only).

100–2—Standard Specification for Marking Shipments by Contractors.[1]

50–11–65—Cloth, Cotton, Cartridge.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

B. GRADE AND CLASSES.

B–1. The dye covered by this specification shall be of one grade and shall be furnished in the following classes of colors as specified in the contract or order. (See par. H–2.)

Class 1—Green.
Class 2—Scarlet.

C. MATERIAL AND WORKMANSHIP.

C–1. See section E.

D. GENERAL REQUIREMENTS.

D–1. See section E.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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E. DETAIL REQUIREMENTS.

E–1. *Form.*—The dye shall be furnished in powdered form, completely free from hard lumps.

E–2. *Uniformity.*—The components of the mixed dye shall be thoroughly and intimately mixed so as to produce a finished dye which is uniform and does not show streaks of different colors when spread out in a thin layer.

E–3. *Applicability.*—The dye shall be of such composition that when applied, as described in paragraph F–4c, to either silk cartridge cloth conforming to the requirements of Joint Army-Navy Specification JAN–C–539 or cotton cartridge cloth conforming to the requirements of U. S. Army Specification 50–11–65, the resulting dyed fabric will acceptably match the specified class of color.

E–4. *Color.*—

E–4a. *Green (class 1).*—The green dye shall produce a distinct light green color.

E–4b. *Scarlet (class 2).*—The scarlet dye shall produce a color closely approximating the color of a Quartermaster standard cloth sample equivalent to red color No. 105 of the color card supplement to U. S. Army Specification 3–1.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Size of lot.*—A lot shall consist of not more than 1,000 pounds of dye of one class.

F–2. *Sampling.*—Select 10 percent, but in no case more than 10 containers, of the containers comprising the lot so as to be representative of the entire lot. From each selected container remove sufficient of the material, by means of a thief, to form a composite sample of approximately 4 ounces when the portions removed from the selected containers are united. Place the 4-ounce composite sample in a suitable glass-stoppered bottle and label so as to show the name of the material, manufacturer, plant, contract or order number, number of pounds in the lot and lot number. Make all acceptance tests on this composite sample.

F–3. *Inspection.*—Unless otherwise specified in the contract or order, inspection shall be at the point of delivery.

F–4. *Tests.*—The laboratory tests shall be made in accordance with the following paragraphs. For Navy purchases, the tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Form.*—The form of the dye shall be determined by visual examination.

F–4b. *Uniformity.*—Place approximately 1 gram of the sample on a piece of white paper and mix well, using a spatula. Press down on the dye with the spatula to break up any lumps or aggregates. The appearance of streaks of different colors indicates nonuniformity due to poorly mixed components.

F–4c. *Applicability and color.*—

F–4c(1). *Silk.*—Cut a sample of silk cartridge cloth, approximately 6 inches wide and 8 inches long, and weigh. Prepare a dye bath of the desired color as follows:

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Water, 50 times the weight of the test piece.
Dye, 2.5 percent of the weight of the test piece.
Sodium sulfate or sodium chloride, 10 percent of the weight of the test piece.

Heat the dye bath to 60°C. and immerse the test piece of cloth in the liquid, agitate well, heat to boiling, and boil for 1 hour. Agitate the bath from time to time to insure that the test piece is totally immersed at all times. Remove the cloth from the bath, wash thoroughly with water and dry for 15 minutes at 100°C. Compare the color of the dyed cloth with that of the specified color to determine if the dye has been applied uniformly and if the color of the test is satisfactory.

F–4c(2). *Cotton.*—Test the suitability of the dye for dyeing cotton cartridge cloth, using the method required for testing the silk cartridge cloth. Agitate the boiling dye bath more frequently to insure uniform dyeing.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packing.*—Unless otherwise specified in the contract or order, the dye shall be delivered in standard 50-pound airtight metal drums so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

G–2. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. *Use.*—The dye covered by this specification is intended for use in dyeing cotton and silk cloth for igniter pads and cartridge bags.

H–2. Requests, requisitions, and contracts or orders should specify the following:

(a) Title, number and date of the specification.
(b) Class of dye required. (See par. B–1.)

H–3. Information relative to Quartermaster standard cloth samples may be obtained upon application to the Office of the Quartermaster General, War Department, Washington 25, D. C.

H–4. Copies of Joint Army-Navy specifications (required for Army and Air Force purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications used by the War Department". Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the Departments of the Army and the Air Force will obtain copies of Joint Army-Navy and United States Army specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–5. Copies of Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy

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Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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